

Quick Check

Lesson 6: Adaptations of the Human Digestive System — Structure Meets Function

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.3: Nutrition in Animals | Lesson 6

A Grade 10 learner studies a diagram of one villus in the small intestine. Inside the villus is a network of tiny blood capillaries very close to the surface cells. What is the main advantage of this capillary network?

- A** It makes mucus to protect the villus from enzymes
- B** It carries absorbed nutrients such as glucose into the bloodstream
- C** It pushes food along the small intestine
- D** It produces amylase to digest the starch from ugali

A learner says: "The small intestine absorbs all the nutrients from ugali, sukuma wiki and nyama choma because it is a very wide tube." The small intestine is actually only about 3 cm wide. Which statement gives the best structural explanation for how it absorbs so much?

- A** Its sphincters hold nutrients in the intestine for several days
- B** Its villi and microvilli (brush border) greatly increase the surface area for absorption
- C** Its strong hydrochloric acid breaks food into particles small enough to pass through
- D** Its wall is made of very thick muscle that squeezes nutrients into the blood

On the Driving Question Board, a group posts a **GREEN** card answering: "How does the stomach avoid digesting itself, even though it makes hydrochloric acid and pepsin?" Which answer is correct and complete?

- A** The stomach wall is covered by a mucus lining, and pepsin is released as inactive pepsinogen
- B** Rugae fold away from the food so that acid never touches the wall
- C** The stomach wall contains no protein, so pepsin cannot harm it
- D** Pepsin works only in the small intestine, so the stomach never contains it

When a learner eats a large meal of ugali and nyama choma, the stomach stretches and churns the food. Which structure of the stomach wall lets it expand and increases contact with the food during churning?

A Villi

B Goblet cells

C Rugae

D Microvilli

The cardiac sphincter is found where the oesophagus joins the stomach. What is its main role in digestion?

- A** It makes pepsinogen for protein digestion
- B** It absorbs glucose from the stomach into the blood
- C** It closes to stop stomach acid and food from flowing back up into the oesophagus (acid reflux)
- D** It controls how quickly chyme leaves the stomach for the small intestine

During the class discussion the teacher asks: "What would happen if the pyloric sphincter opened too quickly and released half-digested chyme into the small intestine before enzymes had finished working?" Which answer shows the correct role of the pyloric sphincter?

- A** It makes the stomach wall fold into rugae
- B** It keeps food in the oesophagus until the stomach is empty
- C** It produces mucus that stops the stomach digesting itself
- D** It controls the release of chyme from the stomach into the small intestine, so digestion is not rushed and can finish

Ugali starch has been digested to glucose by amylase and maltase. Which path correctly shows how this glucose gets into the blood in the small intestine?

- A** It is absorbed by the rugae of the small intestine into the blood
- B** It crosses the microvilli and epithelial cells of a villus into the capillary network inside the villus
- C** It stays in the middle of the intestine and is carried out with the waste
- D** It passes through the pyloric sphincter directly into the blood

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).